



# Introduction to Pandas

BILD 62

# Objectives for today

- Create & manipulate Pandas dataframes
  - Load a tabular file of genetic data into a Pandas dataframe
  - Index and subset Pandas dataframes
  - Generate descriptive statistics for Pandas dataframes
-

Name A Better Trio. I'll Wait 🤔



```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
```

That's it.

[Source](#)

**Pandas** is a useful module that creates “data frames”

- great for real-world, heterogeneous data
- similar to Excel spreadsheets (but way faster!)
- “numpy with labels”
- Smartly deals with missing data

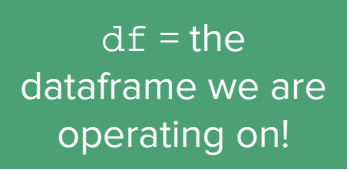
## Numpy:

|   | 0 | 1 | 2 |
|---|---|---|---|
| 0 |   |   |   |
| 1 |   |   |   |
| 2 |   |   |   |

## Pandas:

|          | Height | Weight | Age |
|----------|--------|--------|-----|
| Amy      |        |        |     |
| Brad     |        |        |     |
| Caroline |        |        |     |

df = the  
dataframe we are  
operating on!



## Useful Pandas methods

`df.mean()` Returns the mean of all columns

`df.corr()` Returns the correlation between columns in a data frame

`df.count()` Returns the number of non-null values in each data frame column

`df.max()` Returns the highest value in each column

`df.min()` Returns the lowest value in each column

`df.median()` Returns the median of each column

`df.std()` Returns the standard deviation of each column

For more useful functions, see [this overview](#).

# Resources

[A Quick Introduction to the “Pandas” Python Library](#)

[10 minutes to pandas — pandas 1.5.3 documentation](#)

[Python Data Science with pandas](#)